INCH-POUND
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

DRY CLEANING UNIT ASSEMBLY, FULLY AUTOMATIC, SYNTHETIC SOLVENT, PERCHLOROETHYLENE TYPE, 20-25 POUNDS CAPACITY (NAVAL SHIPBOARD)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

This commercial item description (CID) covers fully automatic dry-cleaning machines of the dry-to-dry, closed solvent recovery type, for using perchloroethylene solvent for Naval shipboard use.

2. SALIENT CHARACTERISTICS

2.1 Design and construction. The complete machine and its components shall be the current standard product of the manufacturer, modified as necessary to meet the requirements of this CID. The machine shall be a self-contained integral unit with:

- combination washer-extractor-dryer
- closed circuit refrigeration-heat pump solvent recovery system
- automatic formula control unit
- still
- filter
- solvent storage tanks
- vapor containment system
- button trap
- water separator
- pumps
- motors
- gauges
- pipes, valves, and fittings

The machine shall be front loading, fully automatic, shell, and open pocket cylinder with a rotating action about the horizontal axis. Rotation of the cylinder shall incorporate reversing action during appropriate parts of the production cycle. The machine shall use chilled water cooling at an entering temperature of 44 °F and shall operate with perchloroethylene cleaning solvent.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: Commander, Naval Sea Systems Command, 2531 Jefferson Davis Highway, Arlington, VA 22242-5160.

AMSC N/A FSC 3510

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2.1.1 Shell. The shell and shell door shall be solvent tight and of sufficient thickness and rigidity to prevent distortion or objectionable vibration under the normal operations onboard ship.

2.1.2 Cylinder. The cylinder shall be rigid to prevent distortion or failure from stresses created during the extract cycle at the rated capacity. Perforations in the cylinder shall be sized and spaced for efficient washing, extraction, and drying. The front of the cylinder shall align with the shell opening in such a way to prevent laundry from falling between the cylinder and the shell. The cylinder shall be provided with lifting ribs equally spaced around the cylinder.

2.1.2.1 Cylinder drive. The cylinder shall be driven by an electric two-speed motor through V-belts or a combination of gearing and V-belts to provide proper speed for the washing, extracting and drying cycles. The cylinder shall reverse rotation not less than three times per 95 seconds during the washing and drying cycles.

2.1.2.2 Cylinder driving trunnion. The cylinder driving trunnion and spider assembly shall be securely bolted to the rear cylinder plate. The shaft opening shall be sealed to prevent solvent leakage during machine operation.

2.1.2.3 Brake. The machine shall be provided with an electric or pneumatic operated brake for rapid deceleration and stopping of the cylinder. The brake shall stop the rotation of a loaded cylinder during any part of the extract cycle within 60 seconds. The temperature rise of the friction surface of the brake lining shall be not greater than 300 °F above the initial temperature of the brake lining when three stops from peak speeds at 2-minute intervals have been made during the extract cycle.

2.1.3 Size and weight. The machine's overall dimensions shall be not greater than 65 inches wide (across front) by 48 inches deep by 76 inches high. The dry weight of the machine including all components shall be not greater than 2750 pounds.

2.1.4 Capacity. The dry weight capacity rating shall be not greater than 3.3 pounds dry weight per cubic foot, or 25 pounds total dry weight.

2.1.5 Mounting. The unit shall be provided with not less than four bolt holes suitable for securing the unit to the deck.

2.1.6 Chilled water cooling lines. Cooling lines shall be configured to provide a smooth flow of water (tight U-bends or similar configurations are prohibited). Separate and independent supply cooling lines shall be provided to the base storage tank, still condenser and refrigeration condenser. The cooling lines shall have no internal connections within the base storage tank, still condenser, or refrigeration condenser. Pressure regulators and pressure gages shall be installed on the inlet side of each cooling water connection. Coils shall be suitable for a working pressure not greater than 45 psi. The machine shall be provided with threaded supply and return connections (of a size suitable for the intended use) to the base storage tank, refrigeration condenser, and still condenser. The flow of water shall be automatically controlled by electric or pneumatic valves. Valves shall be easily accessible for maintenance and replacement.

2.1.7 Steam lines and heating units. The machine shall be provided with threaded supply and return connections to the still and the drying air heating coil. The flow of steam to the drying heating coil shall be automatically controlled by electric thermostatic solenoid valves. A strainer shall be furnished with the steam piping. The still steam heat exchanger shall be a closed-loop type; steam shall not come in contact with solvent. The air

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heating coil shall consist of a continuous tube with closely spaced fins for efficient heat transfer. The air heating coil shall be located in a position in the recovery housing to evenly heat and distribute air entering the cylinder and to efficiently vaporize the solvent and remove and dissipate solvent odors. The heating unit assembly shall be constructed in accordance with ASME Boiler and Pressure Vessel Code, section VIII. Pipes, fittings, valves, coils and chambers using steam shall be suitable for 100 psi saturated steam working pressure.

2.1.8 Automatic valves. Drain valves, solvent inlet valves, steam valves and cooling water valves shall be fully automatic. Valves shall be electrically or pneumatically operated. Thermostatic valves shall be required.

2.1.9 Blower. A two speed blower shall be provided to meet the drying and deodorizing requirements.

2.1.10 Refrigeration recovery circuit.

2.1.10.1 Condenser. The condenser shall be chilled water cooled and of the shell and coil type, the multipass shell and tube type, or the tube-in-tube type. It shall have provisions for cleaning the water circuit without opening the refrigerant circuit. The condenser shall be provided with a water regulating valve. The valve shall be of the direct-acting or pilot-controlled type actuated by condenser gas pressure to modulate the flow of water required for the condenser. The valve shall be selected to regulate from shut off to required capacity within an operating gas pressure rise not greater than 60 psi. The valve shall be adjustable within an operating range of 155 to 225 psi for R-22 refrigerant. The capacity of the water regulating valve shall be not less than 35 psi water pressure. The pressure drop across the valve shall be not greater than 10 psi. The valve shall have flanged connections or union ends. The valve shall prevent the entry of chilled water to the refrigerant system in the event of derangement. The condenser shall use chilled water at an entering temperature of 44 °F. The condenser shall have specific devices for purging both the refrigerant and water circuits and for draining the water circuit.

2.1.10.2 Compressor. The compressor shall be hermetically sealed. The refrigeration system shall be provided with a means of discharging and adding refrigerant to the system. When receivers are used, the compressor shall be provided with suction and discharge valves. The assembled refrigeration system shall be completely dehydrated and charged with the amount of refrigerant and oil necessary for operation. The compressor motor, up to and including 3/4 horsepower shall operate on a nominal 115 VAC source. Compressor motors over 3/4 horsepower shall operate on a nominal 440 VAC source. Motors shall be protected against damage from overloads and voltage fluctuations when starting or running by means of a thermal overload protection and low voltage protection relay.

2.1.10.3 Refrigerant (ozone depleting substance). The refrigerant shall be in accordance with ASHRAE 15, Group A1 (nonflammable), be in accordance with ARI 700 (purity), and be approved by the Environmental Protection Agency under the Significant New Alternate Program. (It shall have an ozone depletion potential of not greater than 0.05.)

2.1.11 Base storage tank. The base storage tank shall consist of two independent solvent tanks. The capacity of each tank shall be not less than 30 gallons. The storage tanks shall be supplied with a closed-loop, in-tank cooling coil to maintain optimum solvent temperature. The base tank cooling coil shall be in accordance with paragraph 2.1.6. A thermostatic valve shall

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be installed on the cooling coil, which opens automatically when the storage tank is above 75 °F and closes when the storage tank is below 60 °F.

2.1.12 Filter. Filtration canisters shall accommodate four paper/carbon core filter cartridges. The filter canisters shall circulate solvent back to the cylinder, draining directly to the still or draining to the base storage tank. Means shall be provided to steam strip filter cartridges within canisters prior to filter replacement.

2.1.13 Vapor containment. Each machine shall be provided with a vapor containment system that prevents the escape of solvent vapors from the cylinder when the cleaned load is being removed. The vapor containment system shall be interlocked with the loading door. When the door is opened, the air in the cylinder basket shall be drawn through slots surrounding the door and into a carbon absorption filtration system.

2.1.14 Still tank. The still holding capacity shall be not less than 30 gallons. The distillation rate shall be not less than 50 gallons per hour. The still shall be fillable directly from the washer shell, filter, or base tank. Steam supply shall be controlled by an automatic solenoid valve that is continuously opened during the distillation process. A float assembly shall be installed in the still that will shut down operation if a preset high level is reached, to prevent solvent over-flow into the condenser and the water separator. A thermostat shall be located on the flow line to the condenser. Steam supply shall be shut off automatically if improper amounts of water are being supplied. The still shall be provided with a drain and clean-out cover to remove residue and oil buildup. An illuminated sight glass shall be provided to allow monitoring of still operation and residue buildup. A pressure relief valve shall be fitted on the still that will release at a pressure of 7 psi.

2.1.15 Solvent level and temperature indicators. The solvent level in the cylinder shall be visible from the front of the machine. The level indicator (or gradation markings) shall be calibrated in inches. High and low levels shall be highlighted. Individual temperature indicators shall be provided for sensing and displaying cylinder air temperature (dry sensor), drying air temperature, evaporator air temperature and solvent inlet temperature. Individual thermostatic control shall be provided for adjusting dry sensor and drying temperature settings. The drying air temperature setting shall be not greater 185 °F.

2.1.16 Interlock and safety devices.

2.1.16.1 Functional modes. Safety and interlocking devices shall function for both clockwise and counterclockwise rotation of the cylinder and when the ship is rolling or listing at not less than 12 degrees from the normal position.

2.1.16.2 Wash-extract interlock. The driving arrangement shall permit changing from the wash to the extract cycle and vice versa with an interlocking feature to ensure smooth transfer.

2.1.16.3 Brake interlock. The main break shall be normally closed and prevent cylinder rotation while the machine is off. Also, in the event of power loss and/or loss of low pressure air the brake shall close automatically.

2.1.16.4 Shell door interlock. The shell door shall be provided with an electric interlock that prevents the door from being opened when the machine is in the extract mode, and prevents rotation of the cylinder when the shell door is opened.

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2.1.16.5 Guards. Exposed belts, shafts, pulleys, and other moving parts shall be fully enclosed by metal guards. Gears shall be fully enclosed in gear housing. Metal guards shall be firmly supported, but shall also be readily removable for maintenance.

2.1.16.6 Vibration switch. The machine shall have an adjustable vibration limit cut-out device to disconnect service if excessive equipment damaging vibration occurs.

2.1.16.7 Drain interlock. An interlock shall be provided which will prevent the machine from extracting until the solvent level has drained to a safe level.

2.1.16.8 Cylinder and blower interlock. The cylinder and blower shall be individually interlocked to prevent the cycling of cylinder heating should the cylinder or blower motor fail.

2.1.16.9 Emergency stop button. An emergency stop button shall be located on the front of the machine within an accessible area. The emergency stop button shall bring the machine to a complete stop when activated. The machine shall reset via a reset button. Activation of the emergency stop button shall be identified by a red indicator light.

2.1.17 Automatic controller. Each machine shall be furnished with an automatic formula control unit for controlling the machine through any complete cycle of fill, wash, drain, extract, dry, deodorize, filter, and distill.

2.1.17.1 Automatic formula control unit. The unit shall be a programmable microprocessor. The automatic formula control unit shall direct the machine through a complete operating cycle as programmed, automatically control temperatures, and produce an audible and visual signal at the end of the cycle or at other programmed times.

2.1.17.2 Control phases. The control unit shall automatically control all phases of clutching and braking, and all motors. The unit shall automatically control the solvent level, which shall be readily adjustable in the field. The control shall automatically add solvent when needed to compensate for solvent absorbed by the load. Drying air temperature and solvent temperature shall be thermostatically regulated by adjusting the flow of steam to the recovery housing and flow of water to base storage tank, respectively. Distillation and filtration shall be automatically controlled by the control unit. Automatic timing of each operation shall only proceed after the proper solvent levels, temperatures, and pressures are present.

2.1.17.3 Microprocessor automatic control unit. The microprocessor automatic formula control unit shall be capable of storing not less than four independent wash formulas. Current wash formulas shall be obtained from Navy Exchange Command, Fleet Assist Team. Preprogrammed formulas shall be filter washed/single bath. The microprocessor shall program distillation and filtration steps during and after the wash cycle. The unit shall be delivered with two formula programs left blank. Preprogrammed formulas shall be retained in static random access memory, in case of power loss to the machine. The back-up source shall be recharged when power is restored to the machine. A locking mechanism shall be provided to prevent accidental reprogramming or tampering by untrained and unauthorized personnel. The circuitry for the microprocessor shall reside on circuit boards which can be removed and replaced using a plug-type receptacle. The mounting of the circuit boards within the microprocessor shall prevent electrical disconnect or shorting under adverse machine vibrations.

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2.1.17.4 Manual control operation. The machines shall have a means to override automatic operation to permit totally independent manual operation for a complete cycle, including filtration and distillation, or interruption during any automatic phase.

2.1.17.5 Simultaneous energizing. The control unit and machine shall be wired so that simultaneous energizing of more than one phase of a formula at a time shall not occur.

2.1.17.6 Pilot light. A digital LED or LCD timer display, indicating time left within the formula, shall be provided on the control unit.

2.1.17.7 Controls, indicator lights, and audible signals. Controls, indicator lights, and audible signals shall be selected and integrated into the machine in accordance with the applicable human engineering requirements in ASTM F1166.

2.2 Material.

2.2.1 Structure. The frame, base, and strength members shall be constructed of malleable iron, ductile iron, or steel that has been hot-dipped galvanized in accordance with ASTM A123. The shell and shell loading door shall be constructed of stainless steel or nickel-copper alloy (not less than 60 percent nickel). Gray cast iron or other brittle materials shall not be used for these parts. Except as otherwise specified herein, gray cast iron may be used for other components such as gear reducers and bearing housings, which are not strength members unless manufactured as an integral part of the frame. Aluminum castings shall not be used.

2.2.2 Shell. The shell, including the shell door, shall be constructed of stainless steel in accordance with ASTM A167, type 302 or 304, nickel-copper alloy (not less than 60 percent nickel) or ferrous metal with stainless steel liners on the side exposed to the washing solution.

2.2.3 Cylinder. The cylinder, driving trunnion, and spider assembly shall be constructed of stainless steel in accordance with ASTM A167, type 302 or 304 or nickel-copper alloy (not less than 60 percent nickel).

2.2.4 Pipes, fittings, valves, and connections.

2.2.4.1 Chilled water cooling lines. Chilled water pipes shall be constructed of copper nickel in accordance with ASTM E75 (copper alloy number 706). Fittings shall be constructed of the same material as piping. Valves shall be cast bronze in accordance with ASTM B61; internal metal parts subject to corrosion or erosion shall be nickel-copper.

2.2.4.2 Steam lines and heating coil. Steam pipes shall be constructed of copper-nickel in accordance with ASTM E75 (copper alloy number 706). Fittings shall be bronze in accordance with ASTM B61 or ASTM B62. Valves shall be cast bronze in accordance with ASTM B61. The drying air steam heating coil shall be made of copper tube with aluminum or copper fins in accordance with ASME Boiler and Pressure Vessel Code, Section II.

2.2.4.3 Solvent lines. Solvent line piping shall be constructed of stainless steel in accordance with ASTM A269. Fittings and valves shall be as specified in paragraph 2.1.8.

2.2.4.4 Air lines. Air line tubing shall be constructed of copper.

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2.2.5 Flexible hoses.

2.2.5.1 Flexible steam hose assemblies. Flexible steam hose assemblies shall be extruded or convoluted tetrafluoroethylene hose with stainless steel wire braid and end fittings in accordance with SAE AS 604 or AS 620.

2.2.5.2 Flexible water hose assemblies. Flexible water hose assemblies shall be Aeroquip Corporation, catalog 306A, model FC254, or model 2580-4 through 2580-32, or equal.

2.2.6 Button trap and water separator. The button trap and water separator shall be constructed of stainless steel in accordance with ASTM A167 or nickel copper alloy (not less than 60 percent nickel).

2.2.7 Refrigeration recovery circuit.

2.2.7.1 Condenser unit. All parts in contact with chilled water shall be constructed of copper-nickel in accordance with ASTM E75 (copper alloy number 706), aluminum bronze in accordance with ASTM B169, ASTM B584 alloy number 1B, ASTM B148 alloy number 9D, ASTM B150 alloy number 2, or valve bronze in accordance with ASTM B61. The condenser chilled water tubes shall be copper-nickel in accordance with ASTM E75 (copper alloy number 706). The water regulating valve body shall be made of valve bronze conforming to ASTM B61. In addition, the internal metal parts subject to corrosion or erosion shall be made of nickel-copper in accordance with ASTM B164. The strainer body shall be valve bronze conforming to ASTM B61, copper-nickel (90-10) or monel. A strainer of number 20 mesh nickel-copper screen of substantial area shall be provided.

2.2.7.2 Evaporator and heating coils. The refrigeration heating and cooling coils shall be constructed of copper tube with aluminum or copper fins. The evaporator coil shall be located in a position in the recovery housing to efficiently condense solvent vapor rising from the cylinder. A thermostatic expansion valve shall be provided to regulate the flow of refrigerant through the coil.

2.2.8 Base storage tank. The base storage tank shall be constructed of heavy-gauge stainless steel in accordance with ASTM A167, or of heavy-gauge steel plate with stainless steel liners on the side exposed to perchloroethylene.

2.2.9 Filter housing. The filter housing and heads shall be constructed of stainless steel in accordance with ASTM A167, or nickel-copper alloy (not less than 60 percent nickel). The filter cartridges shall be paper with a solid carbon core.

2.2.10 Distillation system.

2.2.10.1 Still tank assembly. The still tank shall be constructed of stainless steel in accordance with ASTM A167, or of ferrous metal with stainless steel liners on the interior of the tank. The still heating coil shall be in accordance with paragraph 2.1.7. The tank and coil shall be in accordance with ASME Boiler and Pressure Vessel Code, Section II.

2.2.10.2 Still condenser. The still condenser shall be constructed of stainless steel in accordance with ASTM A167. The still condenser coil shall be in accordance with paragraph 2.1.6.

2.2.11 Recovery housing. The recovery housing shall be made of stainless steel in accordance with ASTM A167.

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2.3 Electrical requirements. The unit shall operate on 440 VAC, 60 Hz, three-phase power, as defined in DOD-STD-1399, Section 300, and shall have provisions for making direct (hardwired) connections for electric power (3 conductors) and for equipment grounding (1 conductor). All outermost surfaces shall be grounded via the equipment grounding connection. The grounding resistance between any exposed metallic surfaces and the common ground point shall be not greater than 0.1 ohm. Electrical components, other than the hermetically sealed motor, shall be provided in accordance with NEMA 250, Type 13 or equivalent enclosure protection. Metal parts of electrical components and enclosures shall be inherently corrosion resistant or shall be treated and processed for corrosion resistance in accordance with IEEE Standard 45. (Shipboard requirement, see 6.2)

2.3.1 Electric motors. Electric motors shall be in accordance with NEMA MG1 and shall have the following characteristics:

Rating	Continuous
Enclosure	Dripproof
Insulation	Class B or F
Voltage	440 VAC, 60 Hz, three-phase
Protection	Thermally protected
Speed Wash/Extract	Two speed, reverse not less than three times per 95 seconds in wash and dry mode
Pump	Single speed
Blower	Two speed

2.3.1.1 Wire. Motors shall be wound with wire sizes in accordance with American Wire Gauge. Half size wire shall not be used.

2.3.1.2 Controllers. Motor controllers shall be of the magnetic type with overload protection relays. Contactors shall switch all three lines of three-phase circuits and both sides of single phase power circuits. Motor controllers shall provide thermal overload protection for motors 1/8-horsepower and larger. Voltage reduction control circuits shall be protected by cartridge fuses sized not greater than 200 percent of the transformer kilovoltampere rating or 100 percent of control wire amperage rating, whichever is smaller. Start and stop pushbuttons and interlocks shall be dripproof. Electrical components mounted outside the control panel shall be provided with dripproof enclosures. Control boxes and electrical or air operated control valves shall be accessible from the front and back sides of the machine.

2.4 Performance requirements.

2.4.1 Environmental suitability. The unit shall be capable of withstanding ship's vibration and motion. Controls, switches, moving parts, and electrical circuits shall operate under shipboard conditions without malfunction, binding, excessive looseness, or damage, when tested in accordance with MIL-STD-167-1, type I equipment. The unit shall be secured to the test machine in the same manner that it will be secured on shipboard. (Shipboard requirement, see 6.2)

2.4.2 Inclined operation. The unit shall operate satisfactorily in accordance with the requirements of this CID (such as the door remaining closed to prevent the loss of contents) with no spillage of liquid or product when test-operated for 30 seconds inclined at an angle of 15° each side of the vertical, in each of two vertical planes at right angles to each other. (Shipboard requirement, see 6.2)

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2.5 Standards compliance. The unit shall be in accordance with and certified to meet the applicable requirements of ASHRAE 15, ARI 700, and UL 969.

2.6 Label plates. The unit shall be provided with a data nameplate and an instruction plate, both attached to the front of the unit. They shall be readily visible during normal operating use and shall not adversely affect the life and utility of the unit.

2.6.1 Data nameplate. The data nameplate shall contain the manufacturer's name, model, serial number, date manufactured, and any other information needed to uniquely identify the unit.

2.6.2 Instruction plate. The instruction plate shall provide instructions for start-up, operation, and shutdown.

2.6.3 Shock hazard labels. A label reading "Danger - Shock Hazard" shall be affixed to the outer case assembly, on or adjacent to each service access cover near one of the fasteners securing the cover. In addition, a warning label in accordance with UL 969 shall be placed near the high voltage components inside the equipment. This label shall include, but not be limited to the following texts:

- Danger - Shock Hazard
- Power supply must be disconnected before servicing.
- Access covers must be in place before use.
- Service should be performed by authorized personnel only.

2.6.4 Harmful vapor labels. A label plate, located for maximum visibility, shall be affixed to the dry cleaning machine. The label plate shall read as follows:

TETRACHLOROETHYLENE
(PERCHLOROETHYLENE)

VAPOR HARMFUL
USE ONLY WITH ADEQUATE VENTILATION

AVOID PROLONGED OR REPEATED
BREATHING OF VAPORS
AND CONTACT WITH SKIN

DO NOT TAKE INTERNALLY

The words, "TETRACHLOROETHYLENE (PERCHLOROETHYLENE)" shall be in yellow lettering, to be legible from not less than 10 feet away, on a black background. The remainder of the label plate shall be in black lettering, not less than 1/8 inch, on a white background.

3. REGULATORY REQUIREMENTS

The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Product conformance. The product provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market, or the same product that has been delivered to the Government for shipboard use on a previous

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procurement. The Government reserves the right to require proof of such compliance.

5. PACKAGING

Preservation, packing, and marking shall be as specified in the contract or purchase order.

6. NOTES

6.1 Ordering data.

- Title, number, and date of this CID
- When required, manuals shall be in accordance with ASTM F760

6.2 Shipboard requirement. Whenever a "(Shipboard requirement)" is included in a paragraph under SALIENT CHARACTERISTICS, it is meant that the requirement is something that is not normally offered to the commercial market by the manufacturer.

6.3 Sources of documents.

6.3.1 Military documents. Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.

DOD-STD-1399	-	Interface Standard for Shipboard Systems Section 300A Electric Power, Alternating
MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)

6.3.2 American Society for Testing and Materials (ASTM) Standards. ASTM Standards are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohoken, PA 19428-2959.

ASTM A167	-	Standard Specification for Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip
ASTM A269	-	Standard Specification for Heat-Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels
ASTM B61	-	Standard Specification for Steam or Valve Bronze Castings
ASTM B62	-	Standard Specification for Composition Bronze or Ounce Metal Castings
ASTM B148	-	Aluminum-Bronze Sand Castings
ASTM B150	-	Aluminum-Bronze Rod, Bar, and Shapes
ASTM B164	-	Nickel-Copper Alloy Rod, Bar, and Wire
ASTM B169	-	Aluminum-Bronze Sheet, Strip, and Rolled Bar
ASTM B584	-	Copper Alloy Sand Castings for General Applications
ASTM E75	-	Chemical Analysis of Copper-Nickel and Copper-Nickel-Zinc Alloys
ASTM F760	-	Food Service Equipment Manuals
ASTM F1166	-	Human Engineering Design for Marine Systems, Equipment, and Facilities

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6.3.3 Society of Automotive Engineers (SAE) Standards. SAE Standards are available from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096-0001.

SAE AS 604 - Hose Assembly, Tetrafluoroethylene, 400 Degrees F, 3000 PSI Hydraulic, Heavyweight
SAE AS 620 - High Temperature Hose Assembly, Convoluted Tetrafluoroethylene, for Aircraft R

6.3.4 National Electrical Manufacturers Association (NEMA) Standards. NEMA Standards are available from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

NEMA MG1 - Motors and Generators
NEMA 250 - Enclosures for Electrical Equipment (1000 Volts Maximum)

6.3.5 American Society of Mechanical Engineers (ASME) Standards. ASME Standards are available from the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

Boiler and Pressure Vessel Code, Section II and VIII - Rules for Construction of Pressure Vessels

6.3.6 American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) Standards. ASHRAE Standards are available from the American Society of Heat, Refrigerating and Air-Conditioning Engineers, 1791 Tullie Circle, N.E., Atlanta, GA 30329-2305.

AHSRAE 15 - Safety Code for Mechanical Refrigeration

6.3.7 Air-Conditioning & Refrigeration Institute (ARI) Standards. ARI Standards are available from the Air-Conditioning & Refrigeration Institute, 4301 North Fairfax Drive, Suite 425, Arlington, VA 22203.

ARI 700 - Specifications for Fluorocarbons and Other Refrigerants

6.3.8 Underwriters Laboratories (UL) Standards. UL Standards are available from the Underwriters Laboratories Inc., 333 Pfingston Road, Northbrook, IL 60062.

UL 969 - Marking and Labeling Systems

6.3.9 The Institute of Electrical and Electronic Engineers (IEEE) Standards. IEEE Standards are available from The Institute of Electrical and Electronic Engineers, 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331.

IEEE 45 - IEEE Recommended Practice for Electric Installations on Shipboard

6.4 Suggested sources of supply. One manufacturer of products known to meet the requirements of this CID is listed below. However, source of supply is not limited to this company, if others are found to meet these requirements of this CID.

Multimatic Corporation
162 Veterans Drive
Northvale, NJ 07647

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MILITARY INTERESTS:

Custodian:
Navy - SH

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA - FSS

Preparing activity:
Navy - SH
(Project 3510-0367)

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